

DATA SCIENCE COURSE · PROJECT PROPOSAL

AI-Driven Water Quality Prediction & Explainable Decision Support System

An end-to-end Data Science workflow: prediction, model comparison, explainability, and decision support.

GROUP 05

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MOTIVATION

Water quality monitoring needs continuous, data-driven insight.

Traditional assessment relies on periodic sampling and lab analysis, limiting continuous monitoring and early detection of contamination events.

Traditional Assessment

- Periodic manual sampling schedules
- Laboratory-based chemical analysis
- Delayed reporting cycles
- Limited spatial and temporal coverage

Data Science Opportunities

- Analyze water-quality patterns
- Predict water-quality conditions
- Identify key influencing parameters
- Support faster, informed decisions

EXISTING APPROACHES

Four major research directions inform this proposal.

Survey scope: recent works (2022–2025) on ML, IoT, WQI prediction, and intelligent monitoring.

01

ML-Based Prediction

Random Forest · XGBoost · SVM · Decision Trees · Regression models

02

Deep Learning

ANN · CNN · LSTM · GRU architectures for temporal and spatial patterns

03

IoT-Based Monitoring

Sensor-based data collection, real-time streams, Cloud / Edge systems

04

Explainable & Intelligent Systems

Explainable AI, anomaly detection, decision-support, pollution prediction

WHAT WE LEARNED

From existing work to a defined project direction.

EXISTING WORK

- Prediction: water-quality classification, WQI prediction, parameter-level prediction
- Monitoring: IoT sensors, real-time data streams, dashboards
- Advanced methods: deep learning, ensemble models, explainable AI

COMMON CHALLENGES

- Limited datasets
- Sensor & data quality issues
- Poor model generalization
- Limited interpretability
- Lack of standardized datasets

OUR OPPORTUNITY

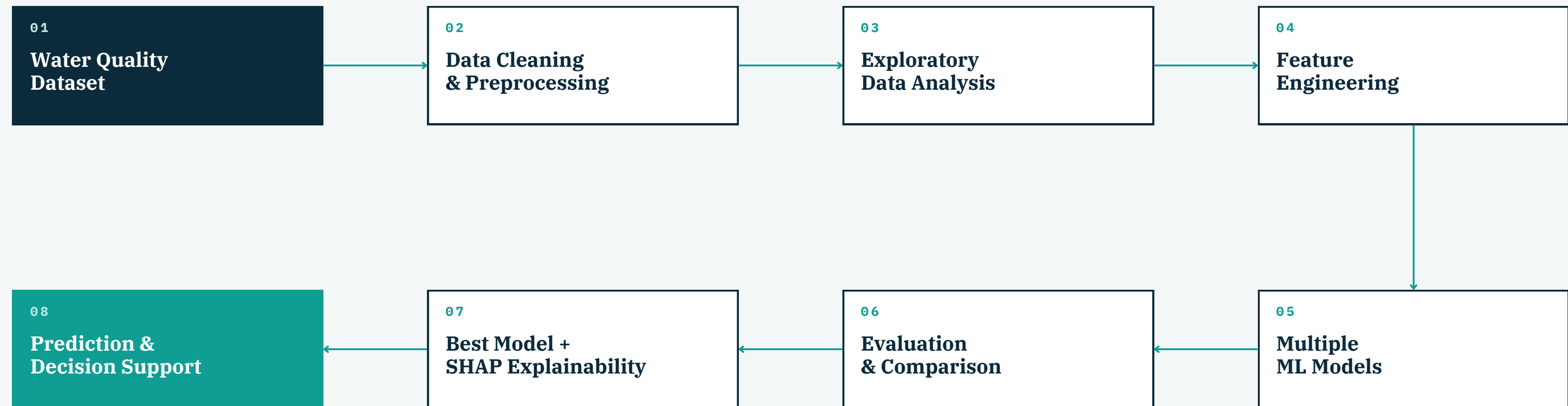
- Develop an end-to-end Data Science workflow combining prediction, model comparison, explainability, and decision support

Framed as project direction, not a claim of novelty.

SYSTEM PIPELINE

An end-to-end prediction and explanation pipeline.

The system predicts water-quality outcomes and provides interpretable explanations for each prediction to support decision-making.



PLANNED METHODOLOGY

Data Science steps, models, and evaluation.

Focus on planned methodology — no results shown at this stage.

A

Data Analysis

- Missing values handling
- Outlier detection
- Correlation analysis
- Feature distribution analysis
- Feature engineering

B

ML Models Compared

- Logistic Regression
- Decision Tree
- Random Forest
- SVM
- XGBoost
- KNN

Selection depends on classification or regression framing.

C

Evaluation Metrics

- Classification: Accuracy, Precision, Recall, F1-score, Confusion Matrix
- Regression / WQI: MAE, RMSE, R²

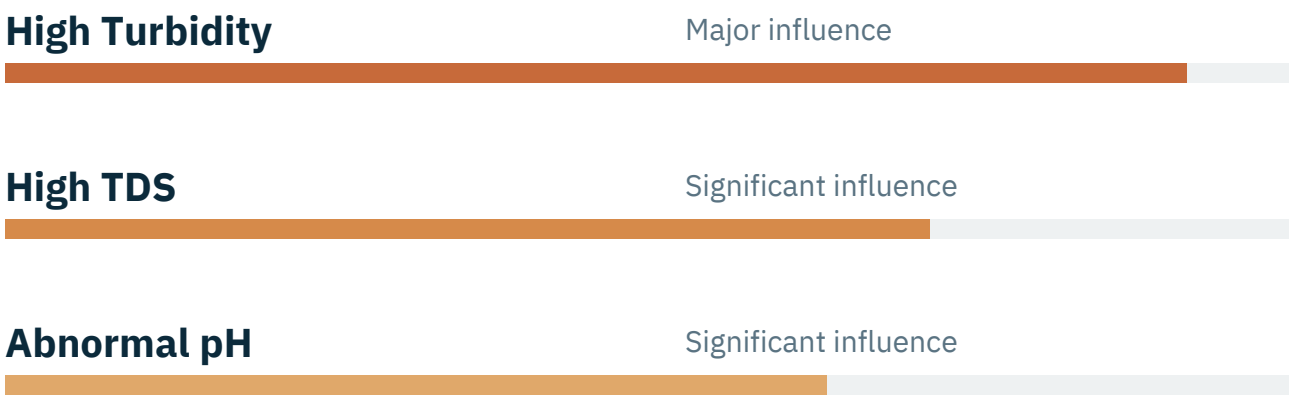
BEYOND PREDICTION

Explainability and an interactive decision dashboard.

ILLUSTRATIVE EXAMPLE – PREDICTION

Poor Water Quality

Top contributing factors (SHAP feature importance):



Illustrative example — actual values will be obtained after model training. Explanations produced via SHAP.

CONCEPTUAL DASHBOARD · STREAMLIT

Decision Support Interface

- Input parameters
- Predicted water quality & confidence
- Parameter visualizations
- Feature importance
- SHAP explanations
- Historical trends visualization

07 · DELIVERABLES & OBJECTIVE

From data to decisions.

EXPECTED DELIVERABLES

- 01 Cleaned and analyzed water-quality dataset
- 02 Exploratory Data Analysis
- 03 Multiple ML models developed
- 04 Comparative performance evaluation
- 05 Best-model selection
- 06 SHAP-based explainability implementation
- 07 Interactive Streamlit dashboard
- 08 Water-quality decision support system prototype

FINAL OBJECTIVE

Develop an end-to-end Data Science application that transforms water-quality data into predictions, interpretable insights, and actionable decision support.

PROPOSED PROJECT

Water Quality Prediction + Explainable AI + Decision Support